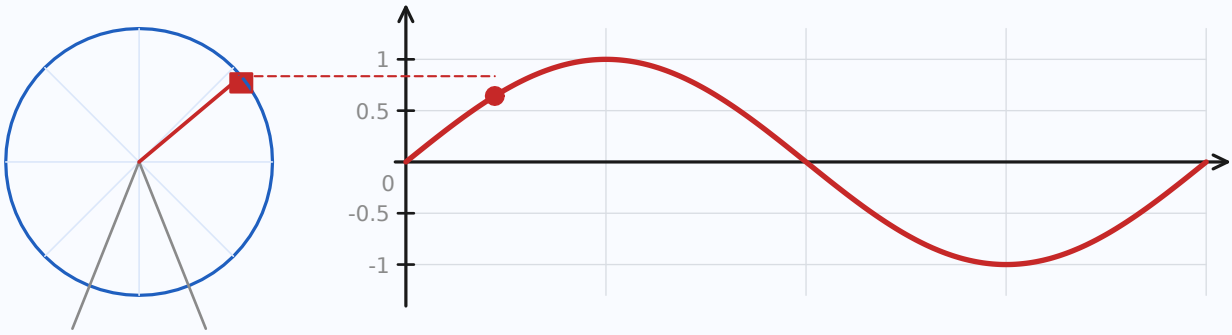
Chapter 29

Trig functions and their graphs

ride the Ferris wheel



your seat's height, turn by turn, draws a wave

In Chapter 28, sin, cos and tan lived inside right-angled triangles, so the angle had to be less than 90°. But angles can be anything: 200°, −45°, ten full turns. This chapter frees sin, cos and tan from the triangle by putting them on a **circle**, and then watches them move.

The star of the show is a Ferris wheel. Ride it, record your height, and you've drawn the sine graph. Everything about the graphs (the repeats, the heights, the shifts) is something about the wheel.

Goes with **Lecture 6** (the unit circle, sin, cos and tan graphs, period, amplitude, power series, odd and even functions).

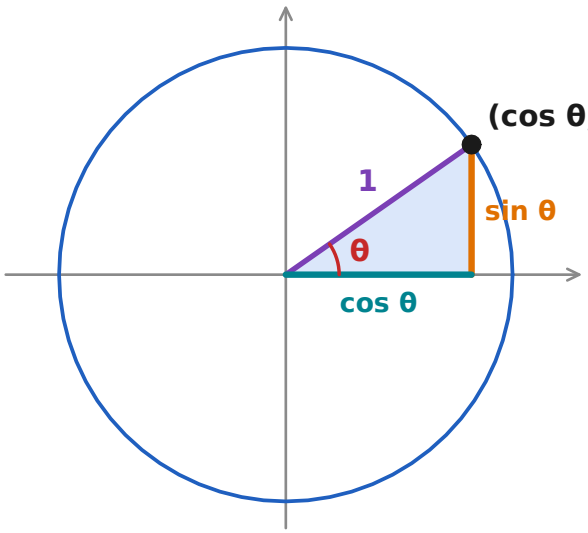
∿ The unit circle: sin and cos become coordinates

RULE: trig of any angle

On the unit circle (radius 1), the point at angle θ is $(\cos\theta,\ \sin\theta)$: **cos is the x-coordinate, sin is the y-coordinate**, and $\tan\theta = \dfrac{\sin\theta}{\cos\theta}$.

This defines sin, cos and tan for **every** angle, not just those under $90°$.

Draw a circle of radius 1 (the **unit circle**, $x^2 + y^2 = 1$ from Chapter 28). Walk anticlockwise from the east by an angle θ and drop a line down to the x-axis. You've made a right-angled triangle with hypotenuse 1:



SOH with hyp = 1: opp = sin θ.
CAH with hyp = 1: adj = cos θ.

So the point on the circle is (cos θ, sin θ) (Lecture 6).

Now let θ keep growing: the point keeps going round, and its x and y define cos and sin for ANY angle.

Signs come free: in the second quarter (90° to 180°) the point is left of the y-axis, so cos is **negative** while sin is still positive. E.g. $\cos 120° = -0.5$, $\sin 120° = 0.866$. And since the point is on $x^2 + y^2 = 1$:

$$\cos^2\theta + \sin^2\theta = 1$$ ⟵ *Pythagoras on the circle (Chapter 30 uses it)*

∿ Any angle: quadrant signs and the reference angle

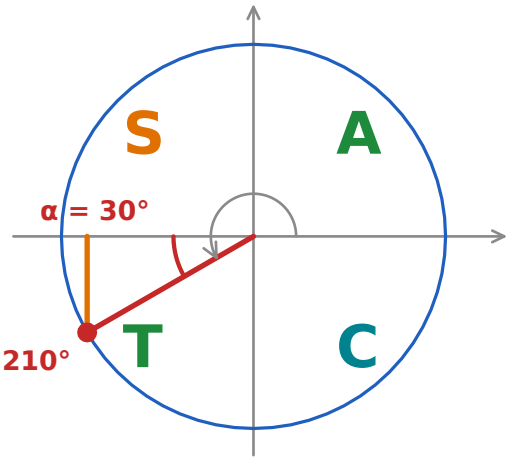
RULE: the quadrant sign rule (CAST)

The four quadrants are numbered anticlockwise from east: Q1 is $0°$–$90°$, Q2 $90°$–$180°$, Q3 $180°$–$270°$, Q4 $270°$–$360°$.

Which ratios are **positive**: Q1 **All**; Q2 only **Sin**; Q3 only **Tan**; Q4 only **Cos**. Memory aid, going round from Q1: "*All Students Take Calculus* ". The rest are negative.

The **reference angle** α is the acute angle between the radius and the **x-axis**. sin, cos and tan of θ are those of α, with the sign from CAST.

Quadrant	θ is between	Positive	Ref. angle α	θ from α
Q1	0° and 90°	all	$\alpha = \theta$	$\theta = \alpha$
Q2	90° and 180°	sin	$\alpha = 180° - \theta$	$180° - \alpha$ $(\pi - \alpha)$
Q3	180° and 270°	tan	$\alpha = \theta - 180°$	$180° + \alpha$ $(\pi + \alpha)$
Q4	270° and 360°	cos	$\alpha = 360° - \theta$	$360° - \alpha$ $(2\pi - \alpha)$



S, A, T, C: the ratios that are positive in each quadrant.

210° is in Q3: the point is below the axis (sin < 0) and left of it (cos < 0). Its reference angle is 210° − 180° = 30°, so sin 210° = −sin 30° = −0.5.

PROCEDURE: sin, cos or tan of any angle

1. Reduce the angle to between $0°$ and $360°$ (add or subtract $360°$).
2. Find its quadrant.
3. Find the reference angle α (table above).
4. Work out sin, cos or tan of α (exact value or calculator).
5. Attach the sign from CAST: $+$ if the ratio is positive in that quadrant, $-$ if not.

$$\sin 210^\circ = -\sin 30^\circ = -\frac{1}{2}$$ ← Q3, α = 30°; sin is −

$$\cos 300^\circ = +\cos 60^\circ = \frac{1}{2}$$ ← Q4, α = 60°; cos is +

$$\tan 135^\circ = -\tan 45^\circ = -1$$ ← Q2, α = 45°; tan is −

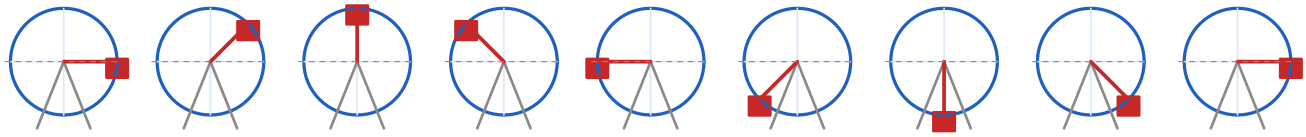
$$\cos(-120^\circ) = \cos 240^\circ = -\cos 60^\circ = -\frac{1}{2}$$

+360°, then Q3, α = 60°

Check on a calculator in DEG mode: $\sin 210^\circ = -0.5$, $\cos 300^\circ = 0.5$, $\tan 135^\circ = -1$. The method matters for exact answers and, in Chapter 30, for running the rule backwards to find every angle with a given sine.

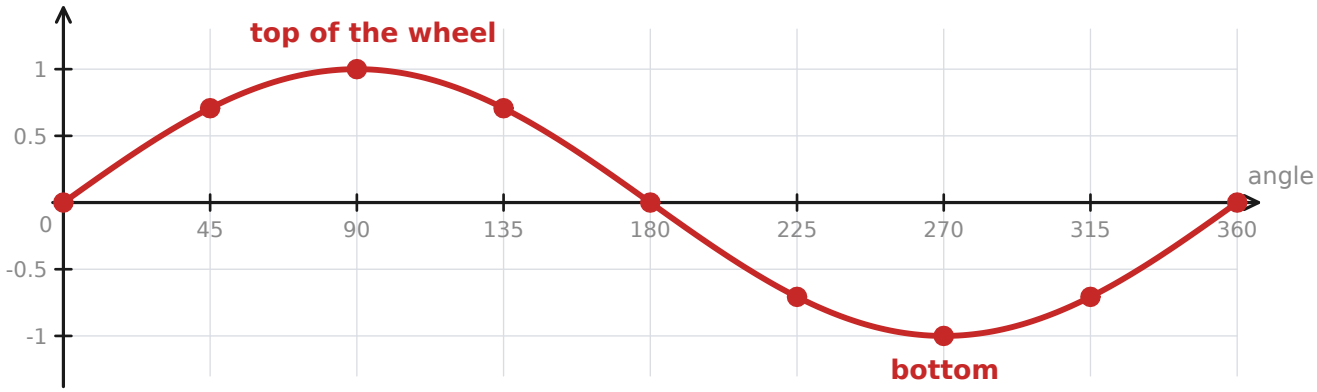
∿ The comic strip: unrolling the wheel

A wheel of radius 1. Start level with the hub (on the right) and turn anticlockwise. In each frame, note how high the seat is **above the hub**:



0°	45°	90°	135°	180°	225°	270°	315°	360°
0	+0.71	+1.00	+0.71	0	-0.71	-1.00	-0.71	0

Plot each height above its angle, then join the dots smoothly:



That wave is the **graph of** $y = \sin\theta$. Keep riding and it repeats every turn, forever (and backwards for negative angles).

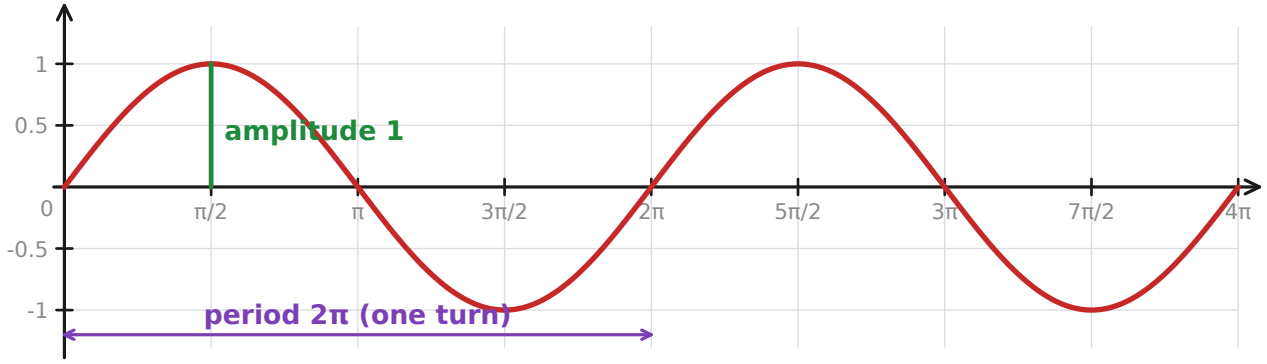
∿ Two numbers describe a wave: period and amplitude

RULE: the sine and cosine graphs

$y = \sin x$ and $y = \cos x$ both have **period** $360°$ (2π) and **amplitude** 1: they stay between -1 and 1.

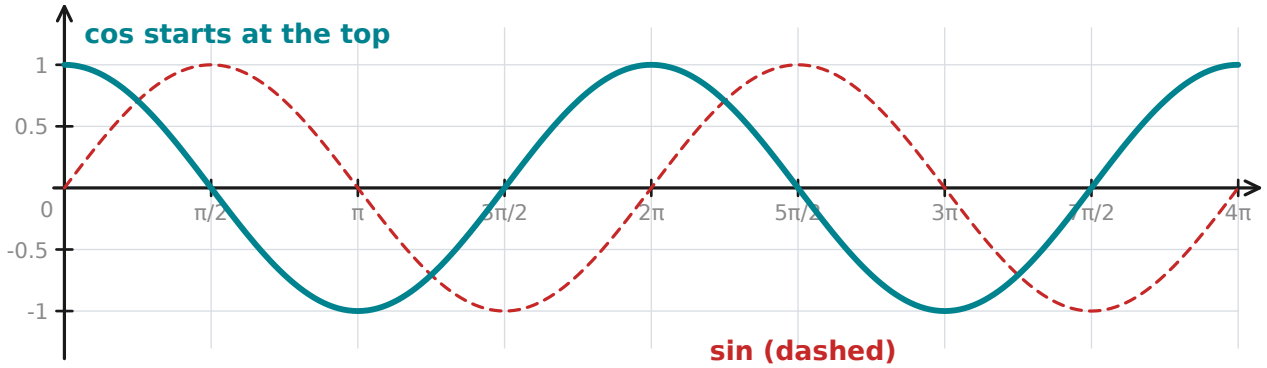
sin starts at 0 going up; cos starts at its peak 1. cos is sin shifted $90°$ left: $\cos x = \sin(x + 90°)$.

cos is even: $\cos(-x) = \cos x$. sin is odd: $\sin(-x) = -\sin x$.



The **period** is how long before it repeats: one full turn, 2π (360°). The **amplitude** is how high it rises above the middle: the wheel's radius, 1. (Lecture 6: sin has period 2π and amplitude 1.)

Cosine is the same ride seen from the side. The seat's **sideways** position (its x-coordinate) starts at 1 (far right), so cos is the sine wave **shifted a quarter turn left**:



Same period (2π) and amplitude (1). Even and odd (Chapter 19): cos is **even**, a mirror image either side of the y-axis ($\cos(-x) = \cos x$); sin is **odd**, it spins onto itself ($\sin(-x) = -\sin x$).

∿ Tan: the wave that explodes

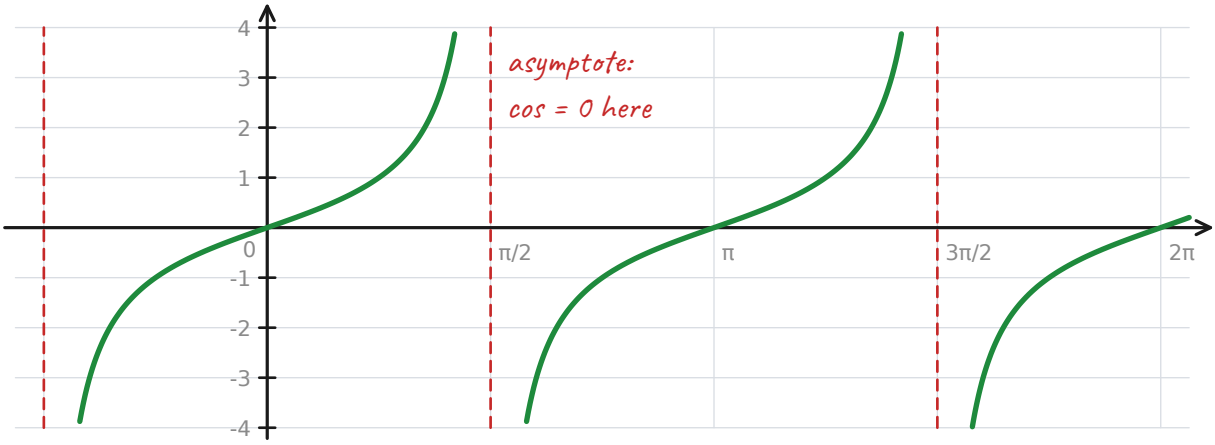
RULE: the tangent graph

$\tan x = \dfrac{\sin x}{\cos x}$. It has **period** $180°$ (π) and no amplitude (no top or bottom).

Vertical **asymptotes** wherever $\cos x = 0$: $x = 90°, 270°, \ldots$ ($\frac{\pi}{2} + k\pi$).

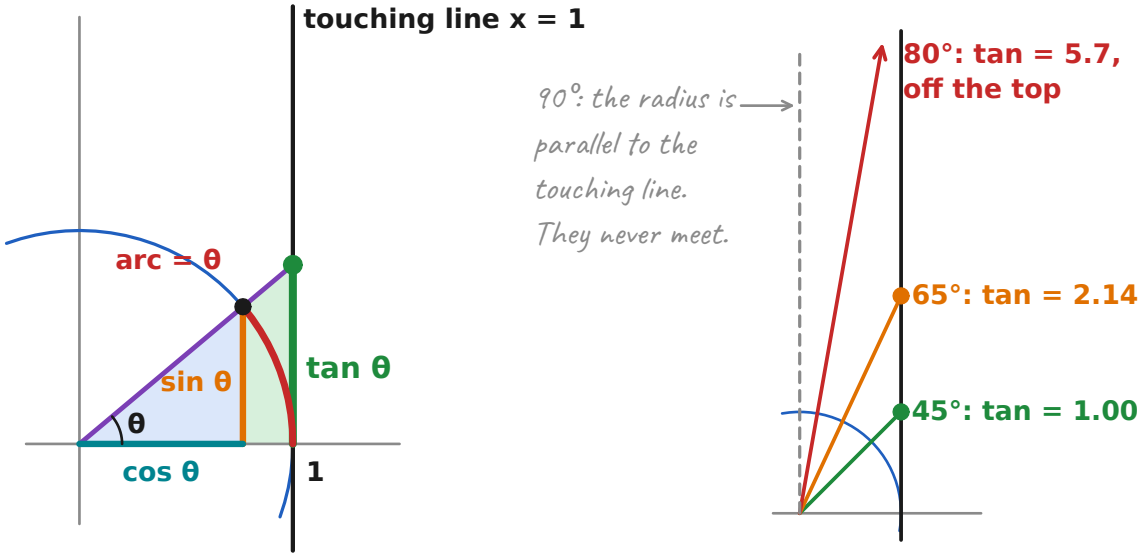
$y = \tan(Bx)$ has period $\dfrac{180°}{B}$, i.e. $\dfrac{\pi}{B}$.

$\tan\theta = \frac{\sin\theta}{\cos\theta}$ (opp/adj = (opp/hyp) ÷ (adj/hyp)). Whenever cos hits 0, at 90° and 270°, we'd be dividing by zero: the graph shoots off to infinity at **vertical asymptotes** (Chapter 19):



Tan repeats every π (180°), not 2π: Lecture 6 says period π and, jokingly, "amplitude of ∞" (it has no top). Tan is odd, like sin.

Why is it called tan? A picture. Draw the line that just **touches** the circle at $(1, 0)$: the upright line $x = 1$. A line that touches a circle is a **tangent** (Latin *tangere*, to touch). Now extend the radius until it hits that line:



The two shaded triangles are the same shape (same angle θ), one scaled up so its bottom is 1. So $\frac{\tan\theta}{1} = \frac{\sin\theta}{\cos\theta}$: **tan θ is the height where the radius line meets the touching line**, which is also the **slope** of the radius (rise ÷ run, Chapter 20). As the angle nears 90° the meeting point races up the line, and at 90° there is no meeting point: that's the asymptote. Past 90°, extend the radius backwards through the centre: it hits the touching line again, which is why tan repeats every half turn.

Notice three heights for the same angle: **sin θ** (shortest), the red **arc θ** (middle) and **tan θ** (tallest). Keep that order in mind; the recipes below will show it.

sin

angle in (the distance walked round the unit circle) → the point's HEIGHT out

cos

angle in → the point's SIDEWAYS position out

tan

angle in → where the radius line hits the touching line x = 1, i.e. the radius's SLOPE (sin ÷ cos)

∿ Bigger wheels, faster wheels

RULE: amplitude, period and shifts

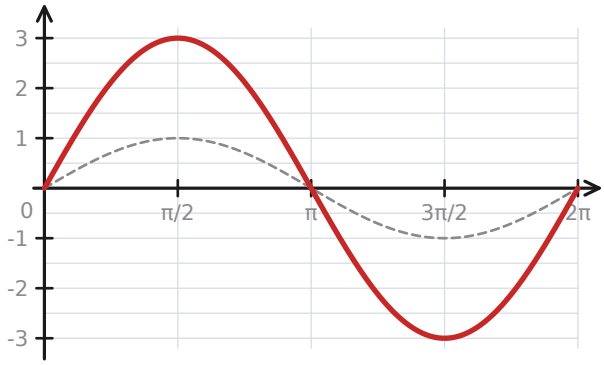
For $y = A\sin\big(B(x - C)\big) + D$ (the same for cos):

Amplitude $= |A|$. **Period** $= \dfrac{360^\circ}{B} = \dfrac{2\pi}{B}$. **Shift** C to the **right** (left if C is negative). **Midline** $y = D$.

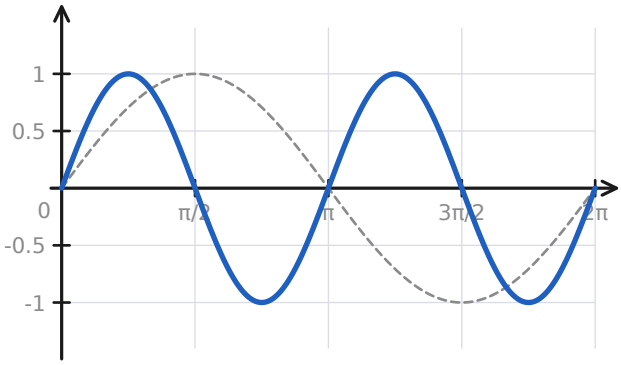
Maximum $= D + |A|$, minimum $= D - |A|$. A negative A flips the wave upside down.

For $y = \tan(Bx)$ the period is $\dfrac{180^\circ}{B} = \dfrac{\pi}{B}$ (tan has no amplitude).

Change the wheel, change the wave. For $y = A\sin(Bx)$:



y = 3 sin x: radius 3, amplitude 3



y = sin 2x: twice as fast, period π

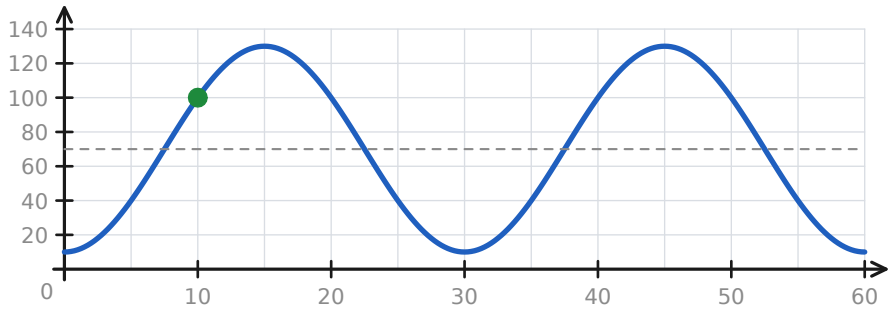
$$\text{amplitude} = A, \qquad \text{period} = \frac{2\pi}{B}$$

Add a number D on the end and the whole wave lifts up by D: the hub is higher off the ground.

The London Eye: radius 60 m, hub 70 m up, one turn every 30 minutes. You board at the **bottom**, so your height starts low: a flipped cosine.

$$h(t) = 70 - 60\cos\left(\frac{2\pi t}{30}\right)$$ ← *t in minutes*

$$h(10) = 70 - 60\cos\left(\tfrac{2\pi}{3}\right) = 70 - 60(-0.5) = 100 \text{ m}$$



minutes along the bottom, metres up the side. Highest (130 m) at 15 min.

Reading a formula: $y = 2\sin\big(3(x - 20°)\big) + 1$.

$$\text{amplitude } 2, \quad \text{period } \frac{360°}{3} = 120°$$ ← *A = 2, B = 3*

$$\text{shift } 20° \text{ right}, \quad \text{midline } y = 1$$ ← *C = 20°, D = 1*

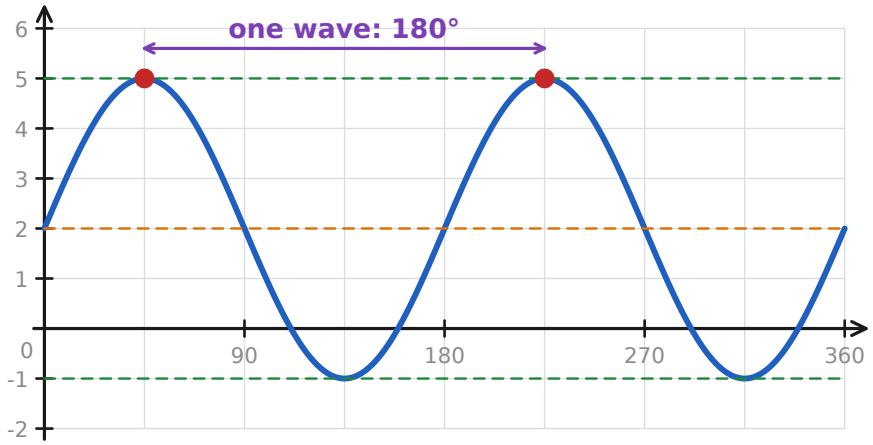
$$\max = 1 + 2 = 3, \quad \min = 1 - 2 = -1$$ ← *D ± |A|*

And $y = \tan 3x$ repeats every $\frac{180°}{3} = 60°$ ($\frac{\pi}{3}$), with asymptotes $60°$ apart.

∿ Reading A, B and D off a graph

PROCEDURE: find A, B and D from a graph of y = A sin(Bx) + D

1. Read the **maximum** and **minimum** heights.
2. Midline $D = \dfrac{\max + \min}{2}$ (halfway). Amplitude $A = \dfrac{\max - \min}{2}$ (half the height).
3. Read the **period**: the distance from one peak to the next (or one full wave).
4. $B = \dfrac{360°}{\text{period}}$ (or $\dfrac{2\pi}{\text{period}}$ in radians).
5. Check one point: at $x = 0$, $y = A\sin 0 + D = D$.



max 5, min −1 (green lines)

orange line: halfway

peaks at 45° and 225°

$$D = \frac{5 + (-1)}{2} = 2, \qquad A = \frac{5 - (-1)}{2} = 3$$

midline, then half the height

$$\text{period} = 225° - 45° = 180° \;\Rightarrow\; B = \frac{360°}{180°} = 2$$

$$y = 3\sin 2x + 2$$ ← check x = 0: y = 2, on the midline ✓

✗ Common mistakes

✗ period of $\sin 2x = 720°$ ✓ period $= \frac{360°}{2} = 180°$

a bigger B squashes the wave: divide by B, don't multiply

✗ $A = \max = 5$ ✓ $A = \frac{5-(-1)}{2} = 3$

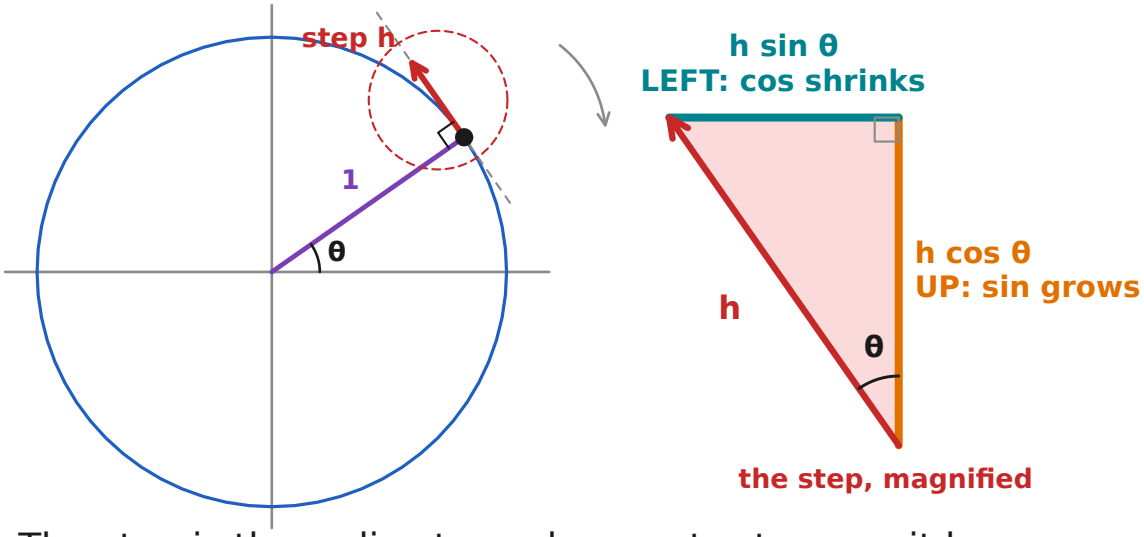
amplitude is measured from the midline, not from zero

∿ The engine inside sin and cos: one tiny step

A calculator can only add, subtract, multiply and divide. So how does it find sin 1? With a recipe made of powers (the **power series**, coming up). The recipe isn't magic: it grows out of one picture of the point walking round the unit circle. Two facts set it up.

1. In radians, the angle IS the distance walked. A radian is one radius of string laid on the rim (Chapter 27). On a circle of radius 1, turning by θ radians moves the point exactly θ along the rim. So a tiny extra turn h moves it a distance h. (In degrees this fails, which is why the recipes need radians.)

2. The point always moves at right angles to the radius, along the line that touches the circle. Zoom in on one tiny step:



The step is the radius turned a quarter turn, so it leans away from upright by the same θ that the radius leans away from flat. Read the little triangle with SOHCAHTOA (hypotenuse h): the point rises by $h\cos\theta$ and slides left by $h\sin\theta$.

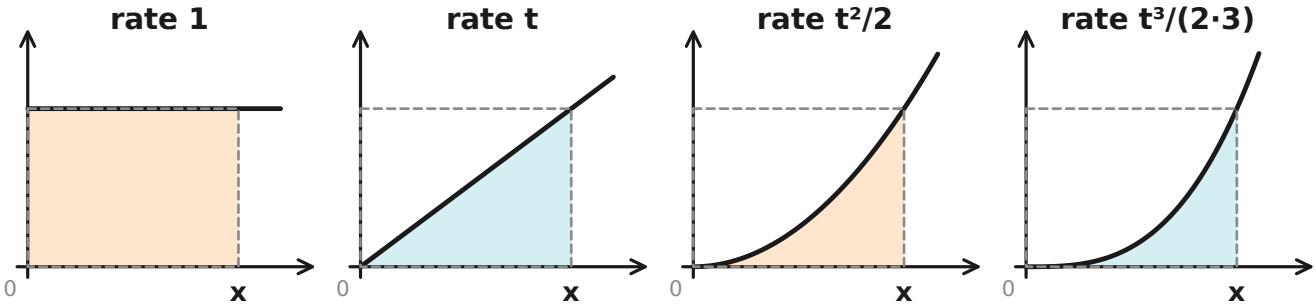
$$\sin(\theta + h) \approx \sin\theta + h\cos\theta \quad \longleftarrow \textit{sin grows at the rate cos says}$$

$$\cos(\theta + h) \approx \cos\theta - h\sin\theta \quad \longleftarrow \textit{cos shrinks at the rate sin says}$$

That's the whole engine: sin grows at cos's rate, cos shrinks at sin's rate. Each one drives the other. Everything else (the waves, the recipes, even Chapter 35's derivatives) comes out of this little triangle.

∿ Building the recipe: leapfrog

Start at angle 0: the point is at $(1, 0)$, so $\sin 0 = 0$ and $\cos 0 = 1$. Now let the angle grow to x and use the engine again and again, each round correcting the last. One tool: when something grows at a known **rate**, its total growth is the **area under the rate graph** (like a speed–time graph, where area = distance). The areas we'll need are all a fraction of a box:



box ÷ 1
$\text{area} = x$

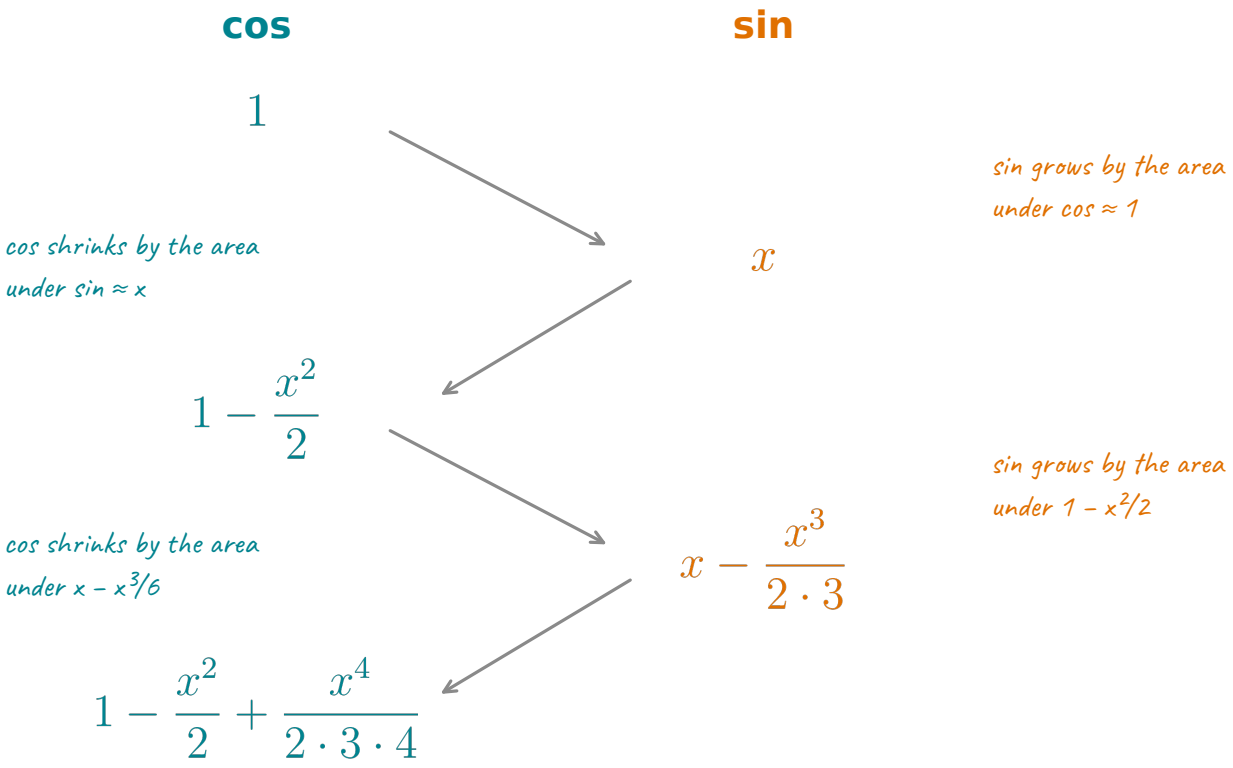
box ÷ 2
$\text{area} = \dfrac{x^2}{2}$

box ÷ 3
$\text{area} = \dfrac{x^3}{2 \cdot 3}$

box ÷ 4
$\text{area} = \dfrac{x^4}{2 \cdot 3 \cdot 4}$

The triangle is obviously half its box. The next two (a third, a quarter) are Chapter 36's pyramid facts; for now, trust the pattern: **each new area is the box ÷ the next whole number**.

Now leapfrog. sin feeds cos, cos feeds sin, and each round uses the latest guess:



Why the signs flip: cos **shrinks** by its area, so every correction to cos is subtracted, and that minus is carried into sin on the next round. Why only odd powers in sin and even ones in cos: each round raises the power by one and swaps sides.

The bottoms are a pattern too: 1, 2, 2×3, 2×3×4, ... Each round divides by the next whole number. That product has a name: $n!$ ("n factorial") means multiply the whole numbers from n down to 1, so $3! = 3 \times 2 \times 1 = 6$ and $5! = 120$. Carry on leapfrogging forever and you get:

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots$$ ← x in radians!

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots$$

And there's the order from the tan picture: $\sin x = x -$ (something), so sin is a little **less** than the arc x.

∿ How does a calculator know sin 1?

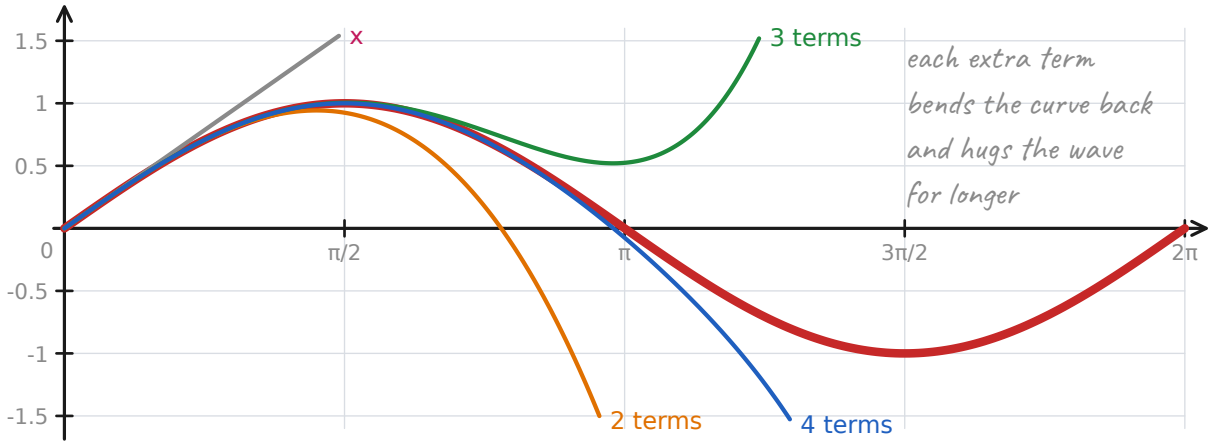
It follows the recipe. Put $x = 1$ and watch the running total home in on the true value:

add 1	1.000000
add − 1/6	0.833333
add + 1/120	0.841667
add − 1/5040	0.841468



Four terms already agree to 5 decimal places: Lecture 6's $\sin(1) \approx 0.841$. Only **odd** powers appear in sin (an odd function) and **even** ones in cos (an even function): that's where the names odd and even come from.

Watch each term earn its place. Here's the real sine wave (thick red) against the recipe cut off after 1, 2, 3 and 4 terms:



x alone rises forever. The $-\frac{x^3}{3!}$ term pulls it back down; $+\frac{x^5}{5!}$ stops it falling too far; and so on. Add enough terms and the recipe matches the wave as far along as you like: the factorials on the bottom grow so fast that they always win in the end.

∿ Tan: one recipe divided by another

$\tan x = \frac{\sin x}{\cos x}$, so divide the recipes. Dividing by $1 - (\text{something small})$ is Chapter 17's sum to infinity read backwards: $\frac{1}{1-r} = 1 + r + r^2 + \dots$

$$\tan x = \frac{x - \frac{x^3}{6} + \dots}{1 - \frac{x^2}{2} + \dots}$$ ← sin recipe ÷ cos recipe

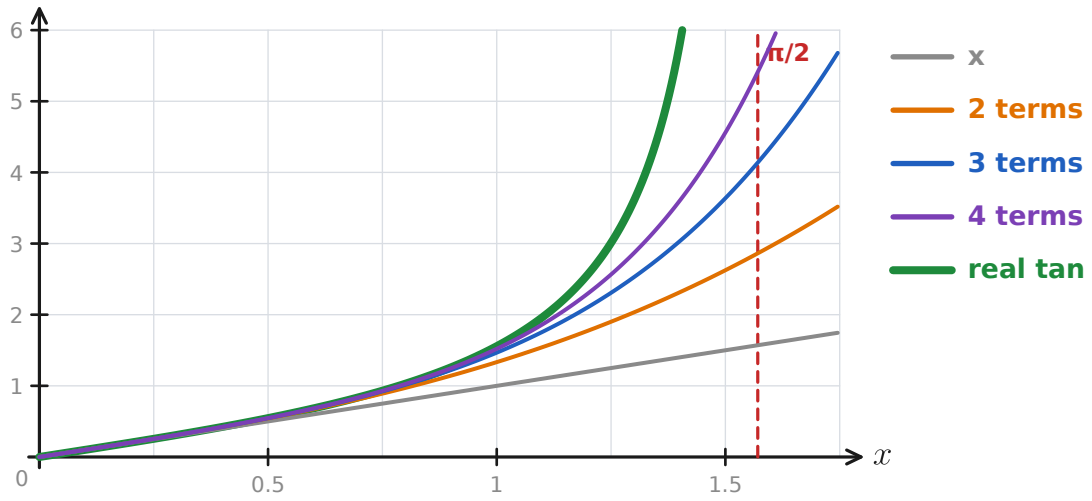
$$\frac{1}{1 - \frac{x^2}{2}} \approx 1 + \frac{x^2}{2}$$ ← Chapter 17 with r = x²/2

$$\tan x \approx \left(x - \frac{x^3}{6}\right)\left(1 + \frac{x^2}{2}\right) = x + \frac{x^3}{2} - \frac{x^3}{6} + \dots$$

expand; keep powers up to x^3

$$\tan x = x + \frac{x^3}{3} + \frac{2x^5}{15} + \frac{17x^7}{315} + \dots$$ ← the rest: by computer

Compare the starts: $\sin x = x - \frac{x^3}{6}$ and $\tan x = x + \frac{x^3}{3}$. sin dips just **below** the arc x, tan climbs **above** it: exactly the order of the three heights in the touching-line picture.



Every recipe is a smooth polynomial, and a polynomial can never shoot off to infinity at a fixed x. So the tan recipe can only work **between $-\frac{\pi}{2}$ and $\frac{\pi}{2}$**. That's Chapter 17's warning in disguise: $\frac{1}{1-r}$ needs r below 1, and here $r = 1 - \cos x$ reaches 1 exactly when $\cos x = 0$, at the asymptote. The division breaks exactly where the picture's radius stops meeting the touching line.

Rules in this chapter

Learn these. Every worked example in the chapter is one of these rules applied step by step.

Unit circle	$(\cos\theta,\ \sin\theta)$, $\tan\theta = \dfrac{\sin\theta}{\cos\theta}$ the point at angle θ
Pythagorean identity	$\cos^2\theta + \sin^2\theta = 1$
CAST	positive: Q1 **All**, Q2 **Sin**, Q3 **Tan**, Q4 **Cos**
Reference angle	Q2: $180° - \theta$; Q3: $\theta - 180°$; Q4: $360° - \theta$
Any angle	reduce to 0–$360°$, find the quadrant, use α, attach the CAST sign
sin and cos graphs	period $360°$ (2π), amplitude 1; cos is even, sin is odd
tan graph	period $180°$ (π); asymptotes where $\cos x = 0$
General wave	$y = A\sin(B(x - C)) + D$ amplitude $\lvert A\rvert$, period $\frac{360°}{B}$, shift C right, midline D; max/min $D \pm \lvert A\rvert$
tan(Bx)	$\text{period} = \dfrac{180°}{B} = \dfrac{\pi}{B}$
Read off a graph	$D = \dfrac{\max + \min}{2}$, $A = \dfrac{\max - \min}{2}$, $B = \dfrac{360°}{\text{period}}$
Power series	$\sin x = x - \dfrac{x^3}{3!} + \dfrac{x^5}{5!} - \dots$, $\cos x = 1 - \dfrac{x^2}{2!} + \dfrac{x^4}{4!} - \dots$ x in radians

∿ Your turn

Answers in grey.

Circle

1. Coordinates of the point at 90° on the unit circle
2. Is cos 150° positive or negative?
3. sin 210° = ? (use the 30° triangle and the sign)

1. (0, 1) 2. negative (left half) 3. −0.5

Graphs

4. Period and amplitude of y = 4 sin x
5. Period of y = sin 3x
6. Period of y = cos(x/2)
7. Where does y = tan x have asymptotes between 0 and 2π?

4. 2π, 4 5. 2π/3 6. 4π 7. x = π/2 and 3π/2

Wheels

8. A wheel of radius 10 m, hub 12 m up, turning once a minute: write h(t) if you board at the bottom.
9. Use the sine series with 3 terms to estimate sin 0.5

8. h = 12 − 10 cos(2πt), t in minutes 9. 0.5 − 0.125/6 + 0.03125/120 ≈ 0.479427

Recipes

10. Use the cosine series with 3 terms to estimate cos 0.5
11. Estimate tan 0.3 with the start of the tan recipe, $x + x^3/3$
12. At 60° (x ≈ 1.047 radians) put sin x, the arc x and tan x in order, with values
13. In the one-tiny-step triangle, which side tells you how fast sin grows?

10. 1 − 0.25/2 + 0.0625/24 ≈ 0.877604 (true 0.877583) 11. 0.3 + 0.027/3 = 0.309 (true 0.3093) 12. sin 0.866 < arc 1.047 < tan 1.732 13. the upright side, h cos θ: sin grows at cos's rate

Find the unknown

14. Exact values of cos 225° and tan 300°

15. θ is in quadrant 2 and sin θ = 0.6. Find θ, cos θ and tan θ.

16. y = A sin(Bx) + D has maximum 7, minimum 1 and repeats every 90°. Find A, B and D.

17. y = 5 sin(Bx) has period 72°. Find B.

18. y = tan(Bx) has asymptotes 45° apart. Find B.

14. $\cos 225° = -\cos 45° = -1/\sqrt{2}$ (Q3); $\tan 300° = -\tan 60° = -\sqrt{3}$ (Q4) 15. $\alpha = \sin^{-1} 0.6 = 36.87°$, $\theta = 180° - 36.87° = 143.13°$; $\cos\theta = -0.8$, $\tan\theta = -0.75$ 16. $D = (7 + 1)/2 = 4$, $A = (7 - 1)/2 = 3$, $B = 360/90 = 4$ 17. $B = 360/72 = 5$ 18. period $45° = 180°/B$, so $B = 4$